

Polytechnic University of Puerto Rico
Biomedical Engineering Department

Spin It 3D

Electrical Subsystem Connection Guide

Clear wiring plan for the MCU, display, sensors, solenoid lock, and motor-speed control.

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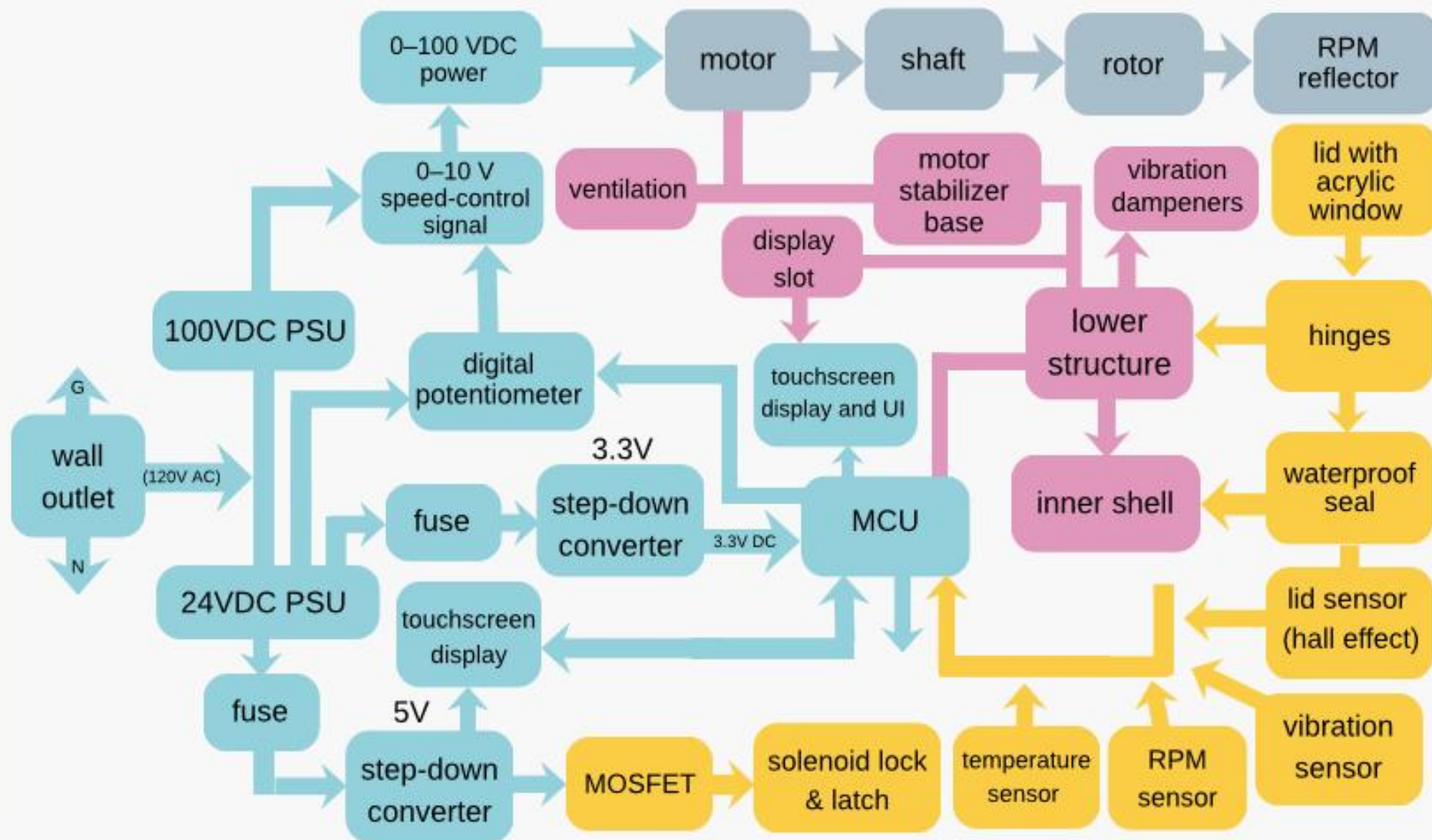
Legend

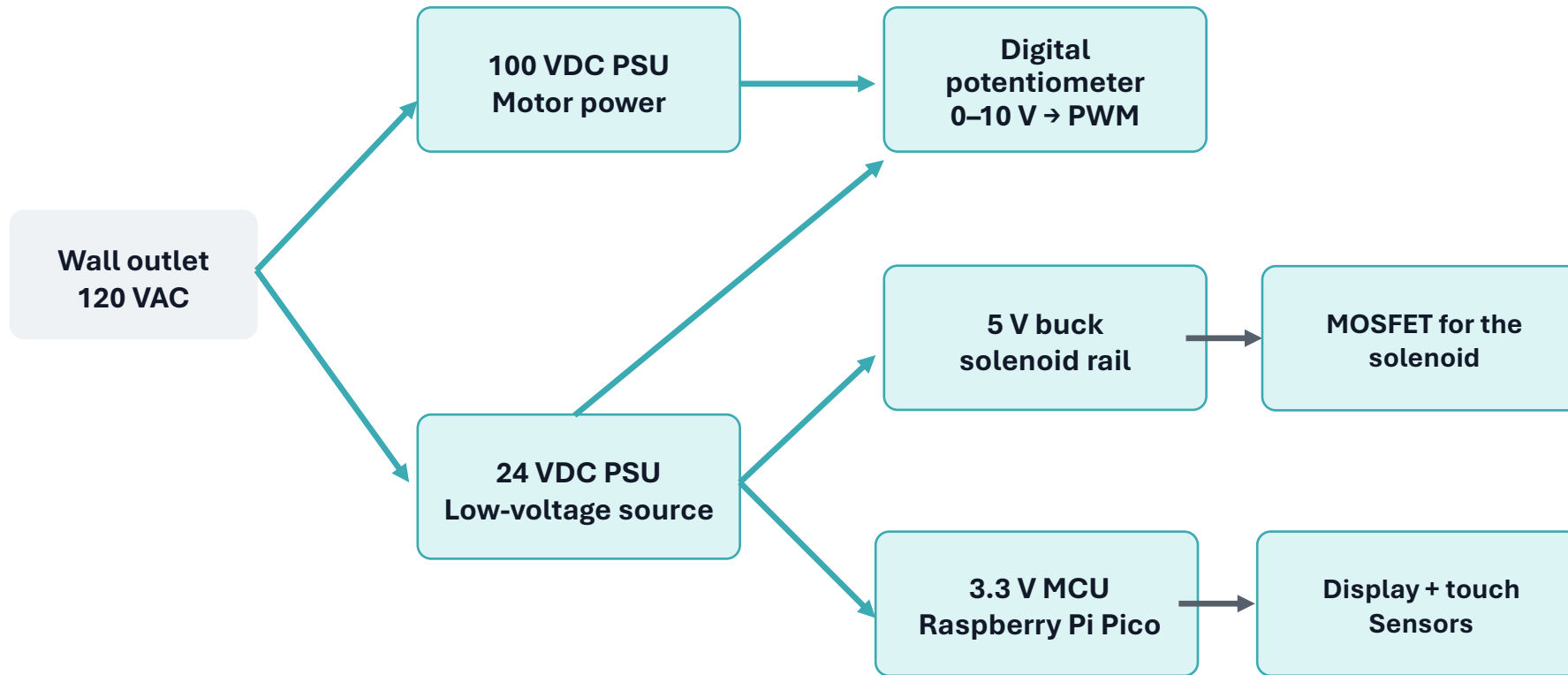
electric
subsystem

rotor
subsystem

lower base
subsystem

lid & sensor
subsystem





Use two power paths: high power for the motor and low power for control electronics.
The MCU commands speed; it should never carry motor current.
All low-voltage modules need a shared ground reference where signals are exchanged.

High-power path

**120 VAC → 100 VDC PSU →
motor and speed controller**

Low-voltage path

**24 VDC PSU → buck converters
→ MCU / solenoid / modules**

Signal reference

**GND must be common for
MCU ↔ sensor/module
signals**

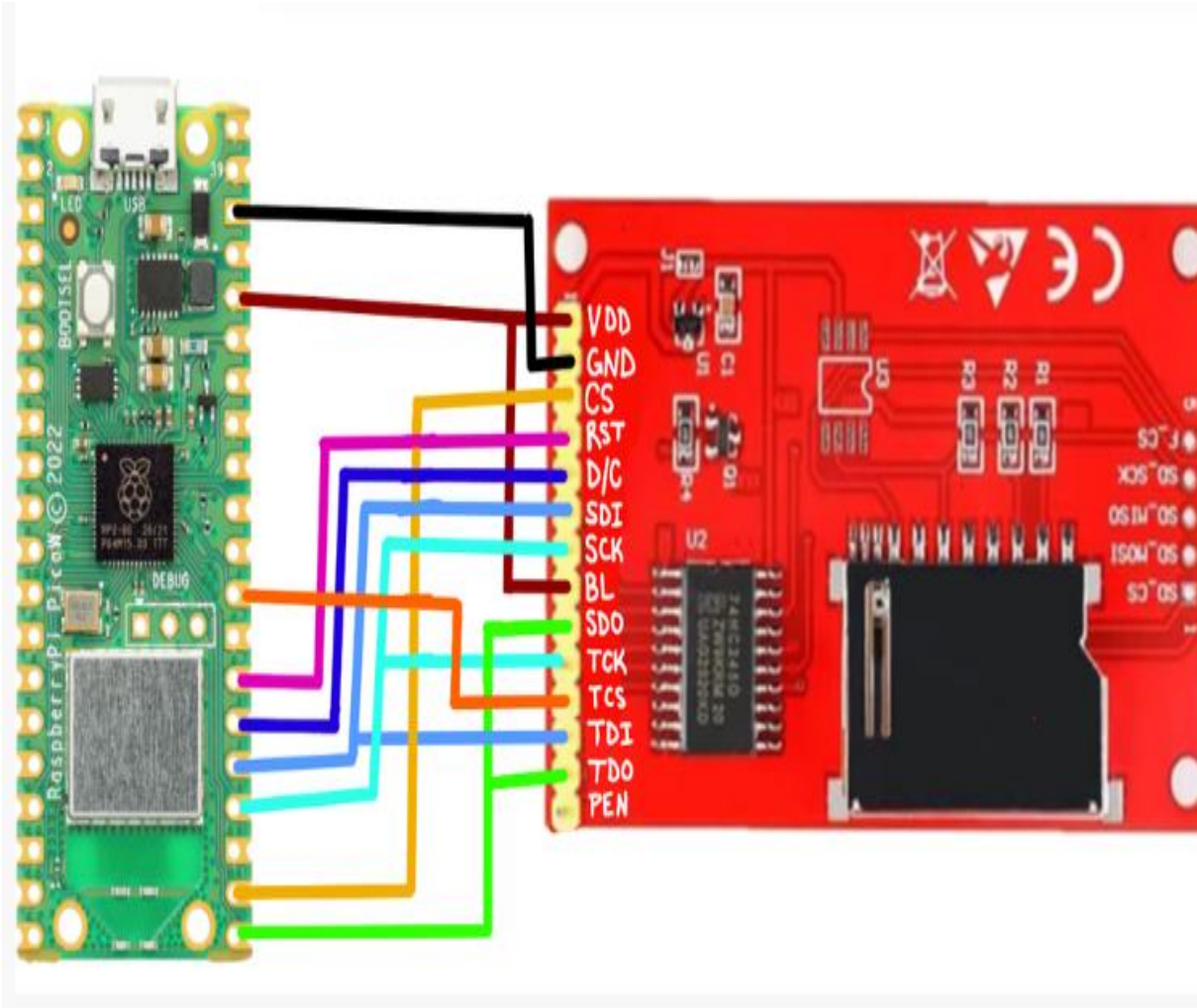
Put a fuse on each supply branch before it feeds the load.

Do not connect the 100 V motor output to the MCU, sensors, display, or breadboard.

Route motor wires away from sensor and touchscreen signal wires to reduce noise.

Use thicker wire for motor and solenoid current; use signal wire only for GPIO lines.

Key idea: power lines deliver current; GPIO lines only send commands.



Use 3.3 V logic

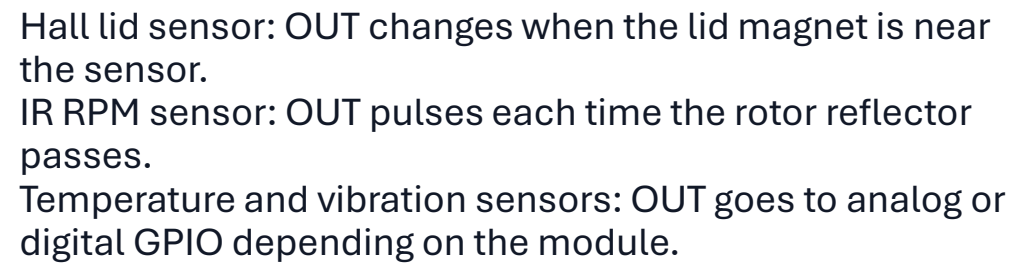
VDD → 3V3, GND → common GND, BL → 3V3 or display backlight control.

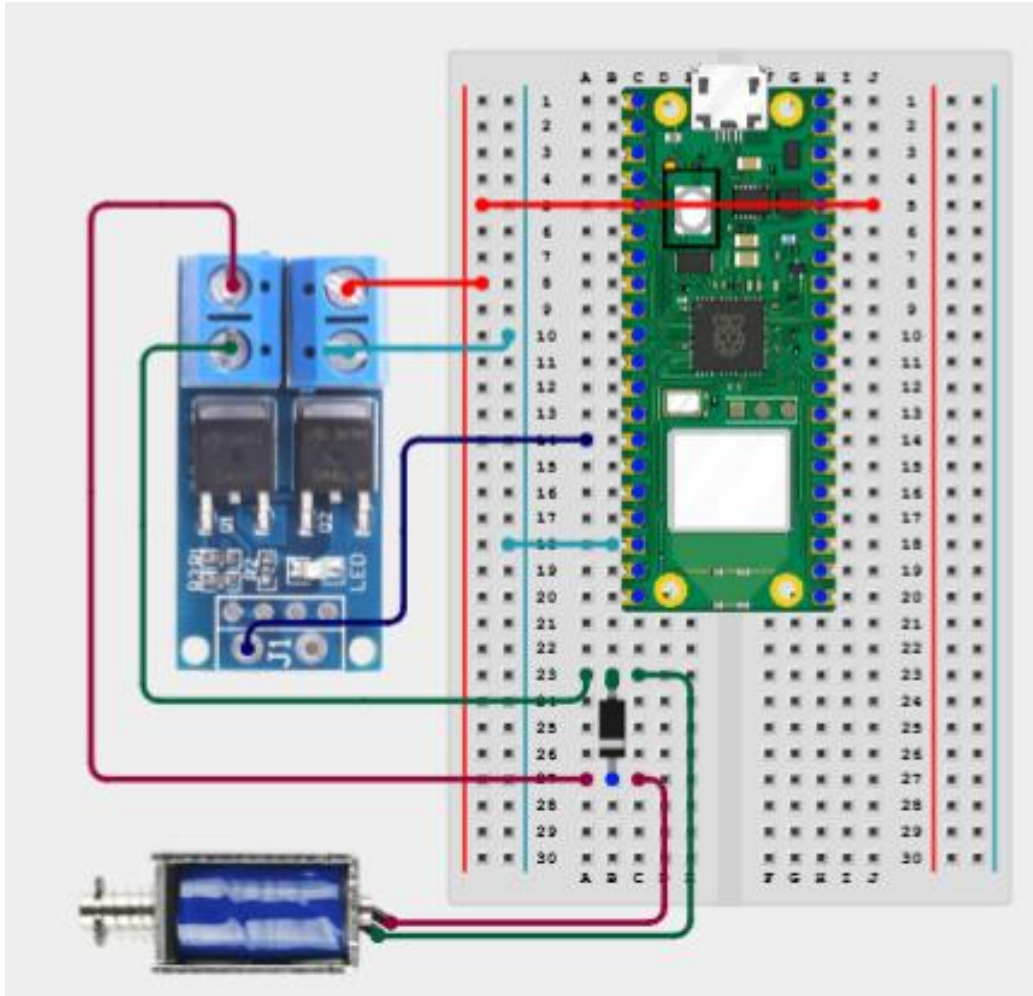
SPI bus: SCK → GP18, SDI/MOSI → GP19, SDO/MISO → GP16.

TFT control: CS → GP17, D/C → GP21, RST → GP20.

Touch control: TCS → GP22; TDI and TDO share the SPI lines; PEN can stay unused.

In code, keep the display pins grouped in one “pin_config.h” file so changes are easy.

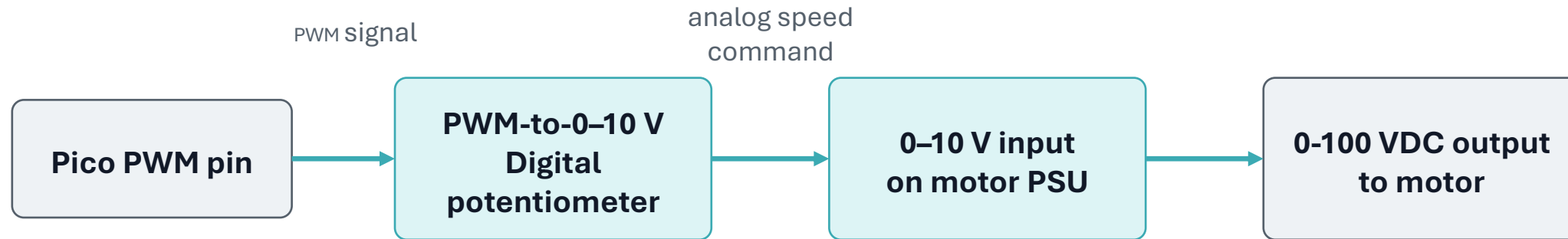




Solenoid wiring

+5 V buck output → solenoid positive lead.
Solenoid negative lead → MOSFET drain.
MOSFET source → 5 V supply and ground.
Flyback diode across solenoid: stripe side to +5 V.

The Pico pin only switches the MOSFET gate. The solenoid current comes from the 5 V buck supply, not from the Pico.



Power the converter from the 24 V rail if the module accepts 12–30 V.
Connect converter OUT+ and GND/OUT– to the PSU’s 0–10 V speed input terminals.
Calibrate the module so 0% PWM $\approx$ 0 V and 100% PWM $\approx$ 10 V.
Keep the 100 V motor output isolated from the MCU and breadboard.

- 1** Verify each power rail with a multimeter before connecting loads and ensure GND continuity.
- 2** Power the Pico + display only; confirm the UI loads correctly.
- 3** Add sensors and check that each GPIO changes state in serial monitor.
- 4** Test the solenoid with a short timed pulse and align it with the latch before installing it on the lid.
- 5** Test motor speed control with no rotor first, then with a balanced rotor and enclosure.

Never test a high-speed rotor without containment, lid interlock, and emergency stop available.